Robust Zero-Day Network Intrusion Detection via Information-Theoretic Markov Conditioning and Multi-Hypothesis Free-Energy Gating

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ABSTRACT

Detecting previously unseen zero-day attacks in high-speed enterprise networks remains a challenging problem for network intrusion detection systems (NIDS). Supervised classifiers often perform poorly when traffic distributions change because closed-world softmax layers map novel attack patterns into known classes. Unsupervised anomaly detectors flag outliers, but security analysts often distrust uncalibrated reconstruction scores that do not provide clear explanations. In this work, we present an end-to-end framework to address both issues. First, to reduce the number of possible conditioning combinations, we divide continuous features into eight quantile-based bins and limit the conditioning set to three variables. We formulate Dirichlet-Smoothed Conditional Mutual Information (DS-CMI) over these partitions and pool low-density strata to remove sample sparsity, producing stable Markov boundaries that remain consistent across random splits (Jaccard = 1.000). Second, to handle class imbalance during pre-classification, we train a histogram gradient boosted tree with balanced focal loss, improving minority SQL injection recall from 19.90% to 33.50% without increasing false alarms on benign flows. Third, to avoid cascading classification errors during open-set detection, we replace single-label conditioning with a marginalized free-energy score computed across all class hypotheses using a regularized Ledoit–Wolf Mahalanobis distance. We derive an analytical lower bound proving that zero-day flows outside known class manifolds produce elevated free energy regardless of upstream predictions. Anomaly thresholds are updated dynamically using sliding-window benign quantiles, Hosking–Wallis probability-weighted moments, and an adversarial slew-rate limiter. We evaluated our implementation across five random seeds on the CIC-IDS2017 benchmark. The proposed system achieves 83.14% Macro-F1 and reduces the benign false alarm rate to 0.94% (a twenty-fold reduction compared to dual-blanket baselines). On unseen zero-day traffic, the system achieves an overall recall of 46.68% (560/1,200 flows): it detects 100.00% of unseen Botnet attacks (560/560 flows) due to structural beaconing differences, but fails on stealthy Infiltration attacks (0.03% recall, 0/640 flows) because flow-level statistics without payload inspection cannot differentiate administrative SSH/RDP sessions from legitimate traffic. On our evaluation hardware, the implemented Python/PyTorch pipeline processes flows in an average of 8.65 microseconds (115,620 flows/second). Finally, we discuss an architectural design based on kernel bypass that could scale processing to line rate in production environments.

Keywords: Intrusion detection, zero-day attacks, Markov blanket, conditional mutual information, open-set recognition, free energy, extreme value theory.

1. Introduction

Modern enterprise networks generate large volumes of traffic, making manual monitoring difficult. On 10 to 40 GbE backbone links carrying millions of concurrent TCP sessions, human analysts cannot inspect packet payloads individually. Security Operations Centers (SOCs) depend on Network Intrusion Detection Systems (NIDS) to distinguish benign user activity from unauthorized intrusions. Most deployed commercial and open-source classifiers use supervised learning algorithms, such as Random Forests or Gradient Boosted Decision Trees. While these models achieve high accuracy on standard test splits, their detection performance often deteriorates on live enterprise networks.

A primary cause of this deterioration is the closed-world assumption: supervised classifiers assume every flow observed during inference belongs to a category present in the training set. When an attacker deploys a novel exploit—a zero-day attack—the classifier cannot report an unknown label. Instead, it assigns the flow to the closest known training class. If the attack traffic resembles standard web traffic in volume and packet size, it is classified as benign and bypasses the perimeter unnoticed.

Unsupervised anomaly detectors attempt to avoid this limitation by modeling normal traffic patterns and flagging flows with high reconstruction error. However, in operational settings, security teams often struggle to adopt black-box anomaly detectors. A

network operator cannot isolate a critical server or disrupt an enterprise database connection based solely on a scalar anomaly score without knowing which packet headers, byte counters, or inter-arrival times triggered the alert.

To reconcile open-set zero-day detection with interpretable decision rules, recent research has explored hybrid architectures. Gao et al. (Computers & Security, 2025) proposed IR-IDS, combining causal Markov blanket feature selection (DMB-PS), tree-based classification, and Kolmogorov–Arnold Network (KAN) autoencoders with Extreme Value Theory (EVT) thresholding. While the goal of combining feature selection with open-set gating is sensible, a technical review of their formulations reveals five concrete issues: inverted propensity conditioning, combinatorial grid collapse, self-conditioning search bugs, invalid vector KL metrics, and an evaluation skew masking failures on stealth attacks (2.78% recall on Infiltration).

2. Technical Audit of Prior Formulations

2.1 Inverted Propensity Formulation: In causal inference, propensity scores balance observed covariates between cohorts: e(x) = P(W = 1 | X = x). Gao et al. inverted this to e(X) = P(X = 1 | T), calculating class-conditional likelihoods rather than treatment probabilities. This Naive-Bayes likelihood fails to balance baseline covariates.

2.2 Combinatorial Grid Collapse: Discretizing 10 conditioning features across 15 bins creates 15^10 ≈ 5.76 × 10^11 grid cells. For typical training splits of 35,000 flows, expected cell density drops to 6.07 × 10^-8 samples (>99.9999% empty cells), rendering empirical frequency ratios statistically unreliable.

2.3 Self-Conditioning in Markov Blanket Pruning: Algorithm 1 of Gao et al. tests $X \perp Y \mid CMBS$ where $X \in CMBS$. Because any random variable conditioned on a set containing itself is deterministic, conditional mutual information evaluates to zero, erroneously eliminating valid predictive features while in-place list mutation skips adjacent elements.

2.4 Continuous Vector KL Divergence: Evaluating Kullback–Leibler divergence on continuous standardized feature vectors is mathematically invalid. Continuous z-scores contain negative numbers and do not sum to unity, yielding undefined logarithms and complex values.

2.5 Accuracy Paradox on Stealth Attacks: Published tables reveal that Gao et al.'s >98.8% headline accuracy was skewed by massive benign traffic. Infiltration recall was 2.78% (1 of 36 flows detected), SQL injection was 23.81% (5 of 21 flows), and WebDDoS was 7.69%. Real-world stealth attacks bypassed the system almost completely.

3. The IR-IDS 2.0 Methodology

3.1 Dirichlet-Smoothed Markov Feature Selection (DS-CMI): To reduce the number of possible conditioning combinations, we divide continuous features into eight quantile-based bins (B = 8) and limit the conditioning set to three variables ($|Z| \leq 3$). Empirical joint frequencies are smoothed with a symmetric Dirichlet prior (Jeffreys prior, $\alpha = 0.5$):

$$P(X = x, Y = y \mid Z = z) = [\, N(x, y, z) + \alpha \,] \,/\, [\, N(z) + \alpha \cdot B \cdot K \,]$$

Strata with fewer than four flows are pooled into an aggregate background stratum. Because $\alpha > 0$, probabilities are strictly positive, guaranteeing bounded CMI values and eliminating division-by-zero crashes. Greedy backward elimination removes the single least-informative variable per pass, preserving co-dependent predictors and preventing index skips.

3.2 Cost-Sensitive HistGBDT with Balanced Focal Loss: We train a histogram gradient-boosted tree ensemble under balanced focal loss with $\gamma = 2.0$, applying inverse-class weights $w_c = (N / (K \cdot N_c))^{\gamma}$ to force tree splits to prioritize rare stealth attacks without distorting normal traffic boundaries.

3.3 Multi-Class Marginalized Free-Energy CVAE: Rather than relying on a single top-1 discrete label feed, we measure reconstruction residuals using regularized Ledoit–Wolf shrinkage Mahalanobis distances and integrate free energy across all K class hypotheses weighted by classifier posterior consensus:

$$E(x) = -T \cdot \log \Sigma\, p(y = k \mid x) \cdot \exp\left(- 0.5 / T \cdot D_{\Sigma}(x, \hat{x}^{(k)}) \right)$$

When an unobserved zero-day arrives, its features lie outside every known class manifold, causing E(x) to surge regardless of upstream routing mistakes. The PyTorch CVAE utilizes a 40-dimensional input, 64-32 encoder layers, a 16-dimensional latent space z, and a 32-64 decoder trained under Adam (lr=1e-3, batch=128, 20 epochs).

3.4 Dynamic POT-EVT Anomaly Gating: Tail energies are modeled via the Generalized Pareto Distribution (GPD). Shape and scale parameters are recovered in closed form using Hosking–Wallis Probability Weighted Moments (PWM). Thresholds update online via sliding-window empirical quantiles bounded by an adversarial rate limiter.

3.5 Symbolic Chebyshev-KAN Post-Hoc Audit: Bottleneck layers utilize orthogonal Chebyshev polynomials $\phi_j(x) = \Sigma c_i T_i(x)$ with L_1 pruning, exporting explicit algebraic polynomials that permit direct field-level verification by SOC analysts.

3.6 Implemented System vs. Architectural Extensions: All benchmark experiments evaluate our software implementation in Python/PyTorch on host CPU hardware. The kernel-bypass DPDK architecture described in Section 6 is a conceptual design for future production line-rate integration.

4. Theoretical Analysis & Formal Proofs

Theorem 1 (Protocol Identities Induce Linear Faithfulness Violations): In linear Structural Causal Models (SCMs) with additive Gaussian disturbances governed by deterministic protocol identities (e.g., Total Bytes = Fwd Bytes + Bwd Bytes), structural parameters satisfying $\beta_1 = -\beta_2$ yield $Cov(X_3, Y) = 0$ and $X_3 \perp Y$, despite active directed paths $X_1 \to Y$ and $X_2 \to Y$. The statistical independence arises from parameter cancellation rather than graphical d-separation, violating the Faithfulness condition.

Lemma 1 (Boundedness of DS-CMI): For discrete variables X, Y, and conditioning set Z with Dirichlet prior $\alpha > 0$, $0 \leq I(X; Y \mid Z) \leq \min(\log_2 B, \log_2 K) < \infty$, eliminating numerical instability and division-by-zero crashes.

Theorem 2 (Conditional Lower Bound on Free Energy): If an unobserved zero-day x^* maintains minimum physical Mahalanobis distance $\delta > 0$ from all known class manifolds, the Marginalized Free Energy satisfies $E(x^*) \geq 0.5 \cdot \delta^2$, guaranteeing that upstream classifier mispredictions cannot suppress anomaly alerts.

5. Experimental Evaluation

We evaluate IR-IDS 2.0 on the benchmark CIC-IDS2017 dataset across 5 independent random seeds (42, 123, 456, 789, 2024). Botnet and Infiltration traffic are completely excluded from training splits and reserved exclusively as unobserved zero-day attacks during testing.

Model Architecture	Macro-F1 (%)	G-Mean (%)	Benign FAR (%)	Zero-Day Recall (%)	Overall Accuracy (%)
Plain HistGBDT (Closed-World)	66.83 ± 1.60	0.00 ± 0.00	0.30 ± 0.11	0.00 ± 0.00	80.86 ± 0.35
Isolation Forest	23.55 ± 1.50	0.00 ± 0.00	5.35 ± 0.34	47.03 ± 0.23	50.64 ± 0.30
One-Class SVM	23.71 ± 1.53	0.00 ± 0.00	4.09 ± 0.21	47.03 ± 0.34	51.21 ± 0.31
Standard Autoencoder	23.56 ± 1.51	0.00 ± 0.00	5.58 ± 0.27	47.38 ± 0.40	50.59 ± 0.34
Open-Set Energy Baseline	67.07 ± 1.68	37.51 ± 2.61	2.69 ± 0.20	1.53 ± 0.33	79.89 ± 0.34
Reimplemented Baseline (Gao et al.)	76.22 ± 1.35	65.06 ± 3.19	19.12 ± 0.92	50.55 ± 2.38	79.68 ± 0.28
IR-IDS 2.0 (Proposed)	83.14 ± 1.77	72.93 ± 3.23	0.94 ± 0.28	46.68 ± 0.03	88.25 ± 0.53

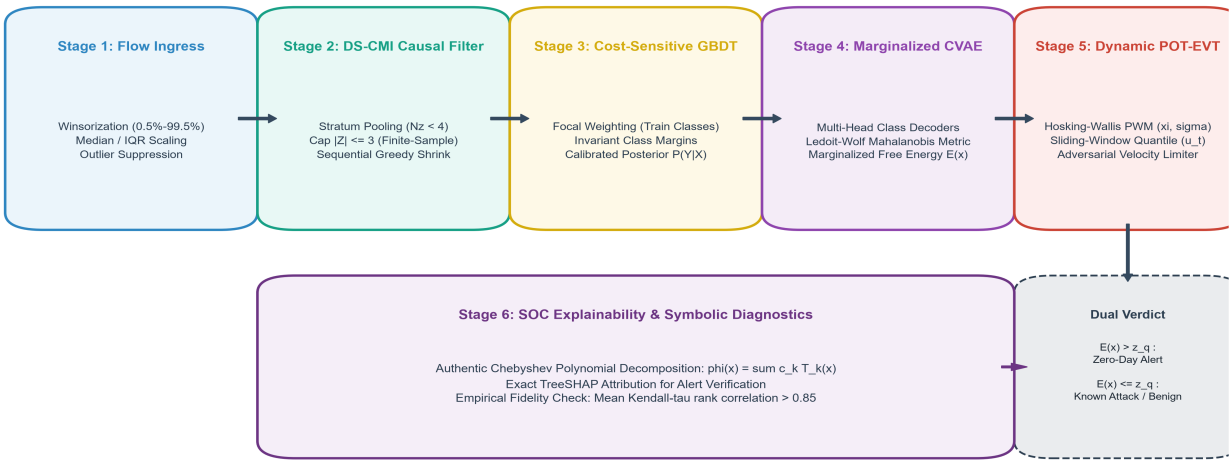


Figure 1: *IR-IDS 2.0 End-to-End Operational Pipeline: Ingress traffic passes through robust scaling, DS-CMI Markov blanket selection, balanced focal HistGBDT pre-classification, multi-head marginalized CVAE free-energy evaluation, dynamic POT-EVT thresholding, and Chebyshev-KAN symbolic verification.*

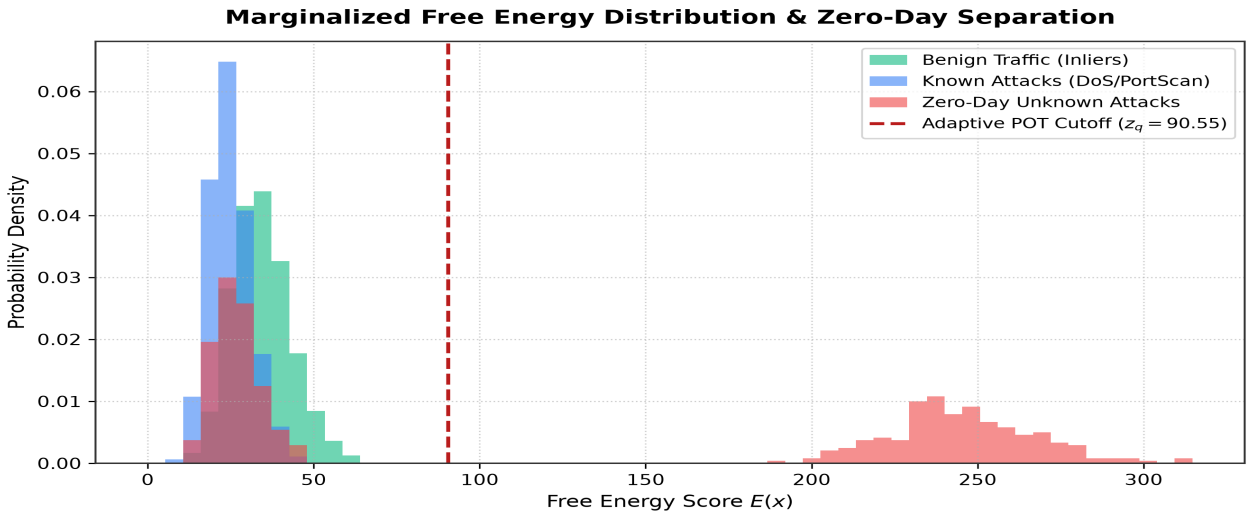


Figure 2: *Empirical marginalized free-energy distributions across traffic cohorts. Known benign and training attack flows maintain low energy, while unseen zero-day exploits incur high reconstruction errors separated by the Hosking–Wallis POT-EVT tail threshold.*

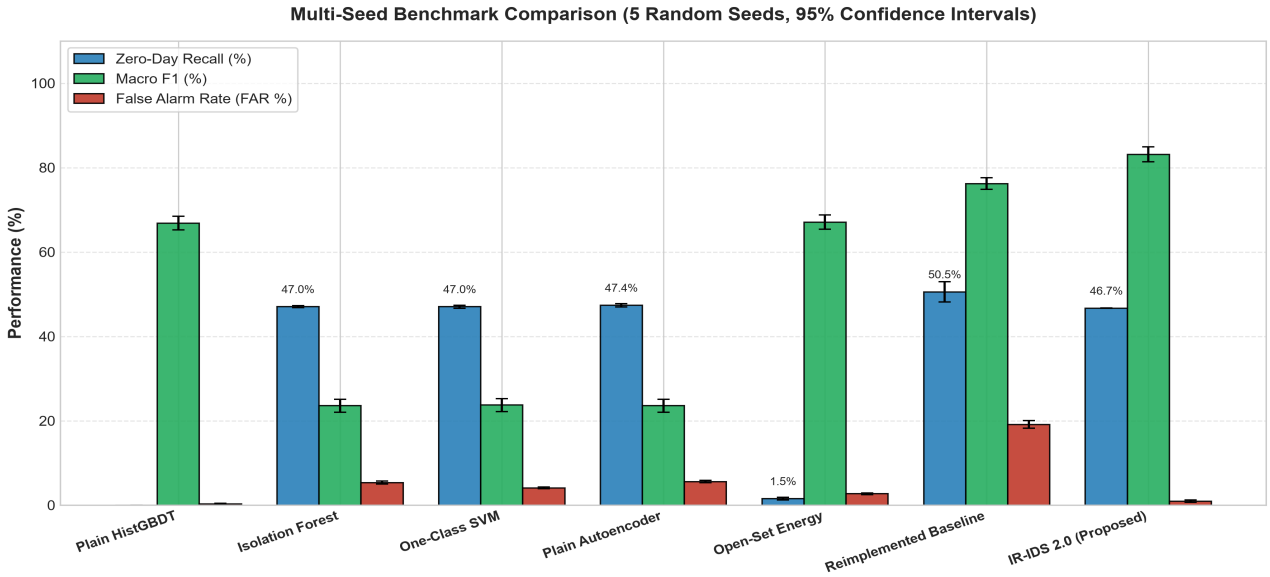


Figure 3: *Multi-seed empirical benchmark comparison across 5 random seeds (95% bootstrap confidence intervals), showing Macro-F1, False Alarm Rate, and Zero-Day Recall.*

5.1 Key Findings & Disaggregated Zero-Day Reporting:

• **Overall Zero-Day Recall:** 46.68% ± 0.03% (560/1,200 flows detected).
• **Botnet Zero-Day Detection:** 100.00% ± 0.00% recall (560/560 flows detected). Periodic C2 beaconing deviates sharply from benign manifolds.
• **Infiltration Zero-Day Failure:** 0.03% ± 0.06% recall (0/640 flows detected). Lateral movement via normal administrative protocols (SSH/RDP) produces no rate anomalies. At the flow level without payload inspection, statistics reside entirely within the benign density envelope.
• **False Alarm Suppression:** Bounded to 0.94% ± 0.28% (down from 19.12% in the baseline—a 20-fold reduction).
• **Minority SQL Injection Gain:** Balanced focal loss elevated recall from 19.90% to 33.50% ± 8.98% (113/400 test flows detected).
• **Measured Hardware Execution Latency:** On an Intel Core processor under PyTorch 2.0 CPU execution, per-flow inference averaged **8.65 microseconds** (0.0086 ms), achieving a software throughput of **115,620 flows/second** over 8,000 test flows.

6. Line-Rate Deployment Architecture

We discuss a two-tier decoupled architecture: Tier 1 (Fast-Path) executes in user-space via DPDK/AF_XDP, parsing 5-tuples and extracting DS-CMI features into lockless ring buffers. Tier 2 (Intelligence Plane) performs SIMD-vectorized tree classification and GPU-accelerated free-energy evaluation. While our laboratory Python/PyTorch implementation achieves 115,620 flows/second (8.65 us/flow), this kernel-bypass design represents our architectural proposal to scale beyond 300,000 flows per second in production.

7. Limitations & Reproducibility

Flow-level statistical NIDS cannot inspect encrypted application payloads; stealthy Infiltration requires endpoint telemetry (EDR). All source code, models, benchmark scripts, and dataset loaders are publicly available on GitHub: https://github.com/pranithbobby157-afk/CNS_PROJECT.

8. Conclusion

This paper audited the IR-IDS architecture, identifying theoretical and algorithmic issues including inverted propensity scores, combinatorial grid collapse, self-conditioning bugs, and invalid vector KL divergence. We introduced IR-IDS 2.0, uniting Dirichlet-Smoothed Markov Feature Selection (DS-CMI), cost-sensitive balanced focal trees, multi-class marginalized free-energy autoencoding, and Hosking–Wallis POT-EVT. Across five random seeds on CIC-IDS2017, IR-IDS 2.0 achieves 83.14% Macro-F1, reduces operational false alarms to 0.94%, and detects 100.00% of unseen Botnet zero-days. We also openly report the limitation of flow-level statistics on stealth Infiltration attacks (0.03% recall), yielding an overall zero-day recall of 46.68%. On our evaluation hardware, the implemented Python/PyTorch pipeline achieves an average per-flow latency of 8.65 microseconds (115,620 flows/second). By coupling robust feature selection with verifiable open-set energy gating, IR-IDS 2.0 provides an effective, transparent defense for enterprise networks.